# NeST-Com: Fixed-Budget Next-Scale Token Communication

***Abstract*—Next-scale token representations support progressive image transmission, but whole-scale endpoints can leave substantial digital-frame capacity unused. We propose NeST-Com, which combines within-scale prefix selection, actual-length fitting, and prefix-constrained reconstruction under a fixed symbol budget. A protected profile specifies the endpoint and modulation/coding scheme, while a shared frozen prior completes missing tokens. With the same prior and image decoder, within-scale endpoints improve PSNR by 0.32–0.70 dB and raise source-bit utilization from approximately 78% to 95% at 10 dB. Five-method experiments characterize the resulting fidelity, perceptual similarity, and processing cost.**

***Index Terms*—Image communication, next-scale prediction, token transmission, generative reconstruction, source–channel coding.**

## I. Introduction

IMAGE communication with a fixed number of channel uses requires the source representation, signaling, and channel protection to share a finite budget. A generative receiver can reconstruct an image from an incomplete representation, making the choice of transmitted information central to reconstruction quality. Next-scale visual autoregressive (VAR) models [1] provide a useful representation for this setting: coarse token maps describe large structures, and successive scales refine their residual information. However, the number of tokens grows with scale, so stopping only after a complete scale can leave substantial capacity unused.

Deep joint source–channel coding (DeepJSCC) maps images directly to channel symbols [2], and SwinJSCC uses hierarchical transformers to improve this mapping [3]. Generative image communication, including DiffCom [4], uses a learned prior to reconstruct images from channel observations. Discrete-token methods provide another route: Ada-TokenCom combines prefix transmission, arithmetic coding, tail generation, and rate/MCS adaptation [5], while TONIC studies token protection and completion for task-oriented communication [6]. Closest to next-scale transmission, ARPC combines scale selection, autoregressive entropy coding, and generation of missing scales for image compression [7]. Its transmitted prefixes end at complete scales, and its compression formulation does not specify a fixed-symbol digital frame.

In a digital link, the relevant constraint is the actual coded length that fits the selected modulation and channel code. This length varies across images, while complete-scale endpoints offer only coarse adjustments. We propose NeST-Com, a communication method that allows the transmitted prefix to end within a scale and fits its terminated arithmetic stream to the available frame capacity. The receiver reconstructs the coding probabilities from decoded coarse scales, preserves the received boundary tokens, and completes the remaining representation. Our same-prior comparison shows that this endpoint flexibility raises source-bit utilization from 77.70% to 95.21% at 10 dB and improves PSNR by 0.32–0.70 dB across the tested SNRs. Comparisons with four existing transmission approaches then characterize the full system's quality and processing cost.

## II. Proposed NeST-Com Framework

### A. System Model and Multiscale Representation

We transmit an RGB image $x \in [0,1]^{256\times256\times3}$ through an additive white Gaussian noise channel,

$$\mathbf{y} = \mathbf{s} + \mathbf{n}, \qquad \mathbf{n} \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}_N), \tag{1}$$

with nominal symbol energy $E_s = 2$ and $\mathrm{SNR} = 10\log_{10}(E_s/\sigma^2)$. The main configuration allocates $N = 1024$ complex channel uses per image. In NeST-Com, 68 symbols carry a protected profile and 956 carry the coded body. The profile identifies the actual source endpoint and modulation and coding scheme (MCS), enabling the receiver to interpret the body.

A frozen residual vector-quantized tokenizer produces token maps $q_1, \ldots, q_S$. We use $S = 10$ scales with side lengths $(1, 2, 3, 4, 5, 6, 8, 10, 13, 16)$ and a 4096-entry codebook. At each scale, the encoder quantizes the residual left by coarser scales. Let $L_s$ be the number of tokens in map $q_s$ and $C_s$ the codebook lookup, resizing, and residual transform that maps this scale to the common latent resolution. The complete representation reconstructs the latent as $f_q = \sum_{s=1}^{S} C_s(q_s)$, followed by the image decoder $D_c$. The tokenizer and VAR prior are frozen; $D_c$ uses adapted weights that are fixed during evaluation.

The frozen VAR predicts the token probabilities of a scale from its complete coarser context:

$$p_\theta(q_s \mid q_{<s}) = \prod_{i=1}^{L_s} p_\theta(q_{s,i} \mid q_{<s}). \tag{2}$$

These probabilities define the arithmetic-coding cumulative distribution functions (CDFs) at both ends of the link. Fig. 1 shows how the representation, digital frame, and receiver completion are connected.

### B. Budget-Adaptive Within-Scale Prefix Transmission

For $m < S$, an endpoint $e = (m, K)$ specifies the transmitted prefix

$$\pi_e = \left(q_1, \ldots, q_m,\, q_{m+1,1:K}\right), \tag{3}$$

where positions within a map follow raster order. $K = 0$ denotes a complete-scale endpoint. For a partial boundary with $L = L_{m+1}$, the implemented candidates are

$$K \in \{\lfloor L/4 \rfloor, \lfloor L/2 \rfloor, \lfloor 3L/4 \rfloor\}. \tag{4}$$

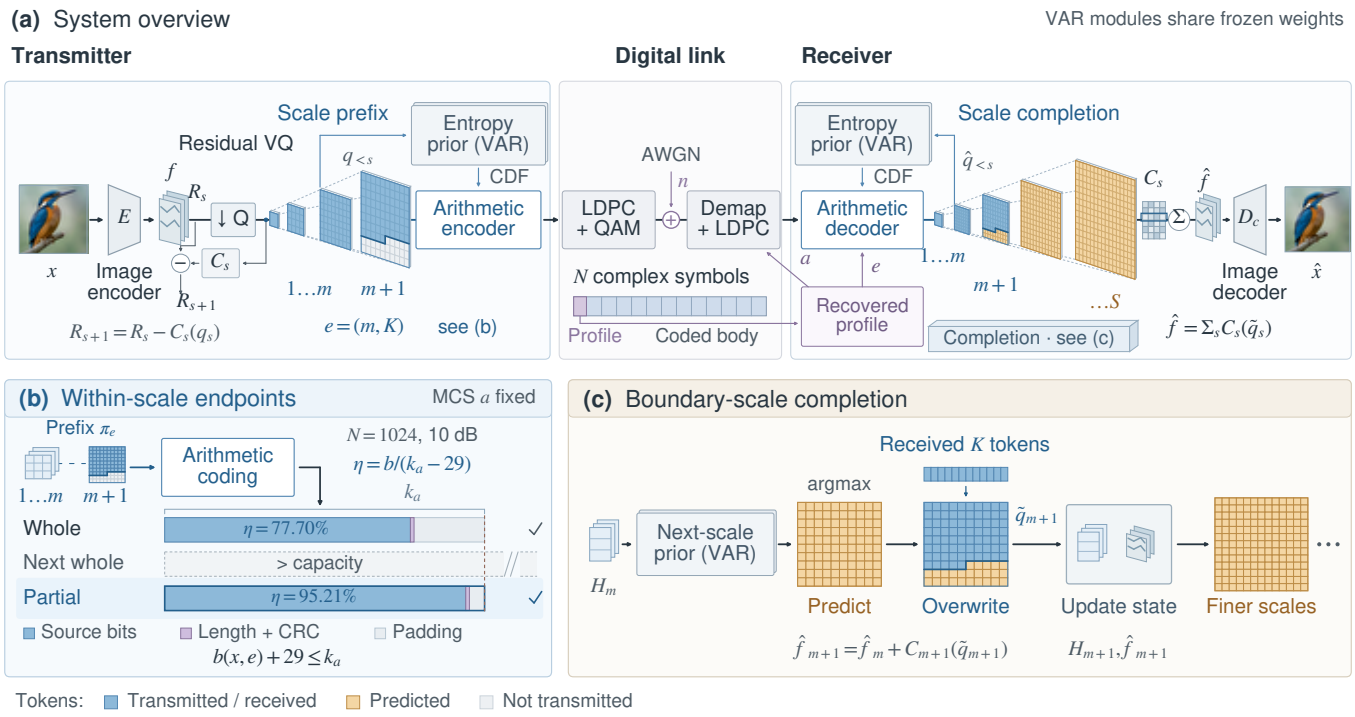



Fig. 1. NeST-Com. (a) The transmitter entropy-codes a next-scale token prefix with endpoint $e = (m, K)$ using CDFs from a frozen VAR. The receiver rebuilds these CDFs from decoded coarse-scale tokens $\hat{q}_{<s}$, and the recovered profile gives $e$ and MCS $a$. (b) The terminated stream of length $b(x, e)$, a 13-bit length field, and CRC16 must fit within the LDPC information length $k_a$. Mean source-bit utilization rises from 77.70% to 95.21% (500 images, $N = 1024$, 10 dB). WHOLE/PARTIAL bar lengths use these measured means. (c) The $K$ received boundary tokens overwrite the VAR predictions; the merged map $\tilde{q}_{m+1}$ conditions finer scales. $H_m$ is the VAR state after scale $m$; $C_s$ comprises codebook lookup, resizing, and a scale-dependent residual transform.

We allow partial endpoints after scales $m = 4, \ldots, 9$. Their resolution is one quarter of a scale, with integer floors for token counts not divisible by four. The complete representation is $e = (S, 0)$ with $\pi_e = (q_1, \ldots, q_S)$ and needs no boundary completion.

For MCS $a = (M, r)$, the 956 body symbols carry $n_a = 956 \log_2 M$ coded bits, and the LDPC information length is $k_a = \lfloor r n_a \rfloor$. Let $b(x, e)$ denote the actual bit length after arithmetic-code termination. A feasible prefix satisfies

$$b(x, e) + 29 \le k_a, \qquad \eta(x, e) = \frac{b(x, e)}{k_a - 29}, \tag{5}$$

where 29 bits comprise a 13-bit stream-length field and a 16-bit cyclic redundancy check (CRC). The remaining information positions are zero-padded before LDPC encoding and quadrature amplitude modulation (QAM).

Offline calibration selects an SNR-specific target endpoint, MCS, and ordered fallback chain. For each image, the transmitter arithmetic-codes the target prefix and moves to earlier legal endpoints until (5) holds, keeping the selected MCS fixed. The protected profile records the resulting endpoint. Checking the terminated length accounts for both image-dependent entropy and coding overhead. Allowing quarter-scale endpoints gives the fitting procedure intermediate choices when the next whole scale exceeds capacity.

We report the source-wise mean of $\eta$. At 10 dB, $k_a = 2549$ leaves 2520 source bits; complete-scale and within-scale endpoint sets use 77.70% and 95.21% of this capacity, respectively. The finite endpoint spacing still leaves some padding, as illustrated in Fig. 1(b).

### C. Prefix-Constrained Reconstruction

After recovering the profile and validating the decoded body, the receiver arithmetic-decodes $\pi_e$. Before decoding each transmitted scale, it reconstructs the CDFs from the already decoded complete coarser maps $\hat{q}_{<s}$. This reproduces the transmitter's probability model without transmitting CDFs.

At the boundary scale, the VAR predicts a full map from $\hat{q}_{\le m}$. Denote its deterministic argmax prediction by $q^{\text{pred}}_{m+1}$. The receiver forms

$$\tilde{q}_{m+1,i} = \begin{cases} \hat{q}_{m+1,i}, & i \le K, \\ q^{\text{pred}}_{m+1,i}, & i > K. \end{cases} \tag{6}$$

The merged map updates the accumulated latent and VAR state before prediction advances to the next scale. Consequently, every received token contributes directly to reconstruction and conditions the finer scales. For complete received scales, $\tilde{q}_s = \hat{q}_s$; at the boundary, it follows (6); for untransmitted finer scales, it is the argmax prediction. The output is

$$\hat{x} = D_c\left(\sum_{s=1}^{S} C_s(\tilde{q}_s)\right). \tag{7}$$

This predict–overwrite–advance procedure makes a partial map a valid context for the same next-scale prior used by the entropy coder.

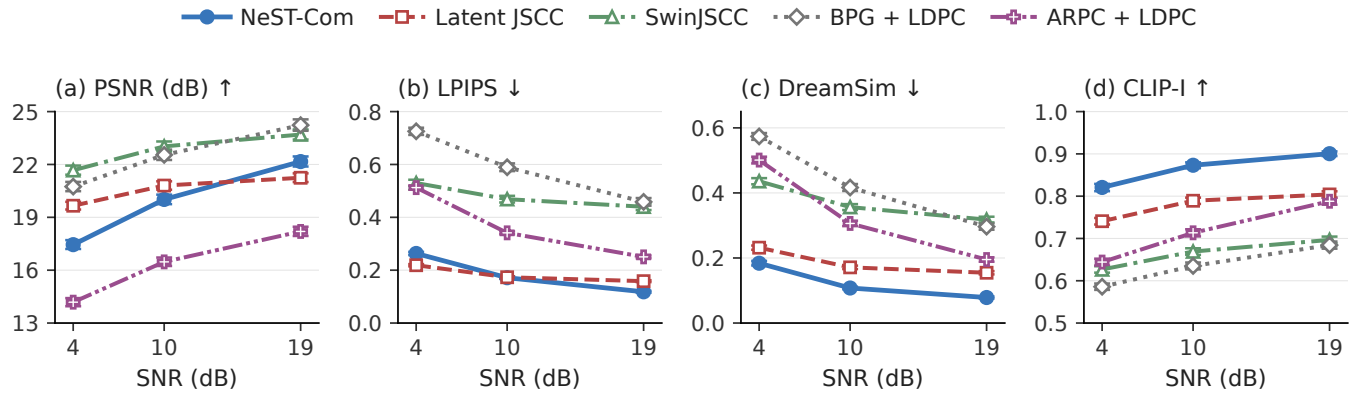



Fig. 2. Five-method performance on common500 at $N = 1024$. Markers and error bars show source means and pointwise 95% bootstrap intervals after averaging three channel realizations per image. Higher PSNR/CLIP-I and lower LPIPS-Alex/DreamSim are favorable.

## III. Experimental Results

### A. Experimental Setup

The main evaluation uses 500 ImageNet validation images [8], denoted common500, at $256 \times 256$ resolution, with $N = 1024$ and SNRs of 4, 10, and 19 dB. All methods use the same total symbol budget and nominal $E_s = 2$. A separate set of 1000 calibration images selects the offline policy by maximizing mean DINOv2 ViT-L/14 image-feature cosine similarity [9].

We compare NeST-Com with Latent JSCC, SwinJSCC [3], BPG + LDPC, and ARPC + LDPC. Latent JSCC transmits a continuous latent through a learned channel mapping, while SwinJSCC directly maps image features to channel symbols. Both JSCC baselines are trained on the same 20 000 images, and Latent JSCC shares $D_c$ with NeST-Com. We use the 100k-update Latent JSCC and adapted 200k-update SwinJSCC checkpoints. BPG [10] selects its resolution and quantization parameter to fit the digital payload. ARPC uses its native scale presets and the authors' frozen weights; its caption is counted in the transmitted payload. Each implementation includes its required signaling in the symbol budget.

We measure pixel fidelity with PSNR, perceptual similarity with LPIPS-Alex [11] and DreamSim [12], and image-feature similarity with CLIP-I, the image–image cosine of CLIP ViT-L/14 embeddings [13]. DreamSim and CLIP use fixed $224 \times 224$ preprocessing and FP32 inference. For every source, we average three channel realizations before computing the source mean; rejected digital receptions contribute the protocol's fixed gray reconstruction. Error bars are pointwise 95% source-bootstrap confidence intervals (CIs), and differences use source-paired intervals. A matched 100-image subset supports the larger-budget and ARPC sensitivity experiments.

### B. Comparison With Existing Methods

Fig. 2 shows that NeST-Com attains the lowest DreamSim and highest CLIP-I means at all three tested SNRs. At 10 dB, its DreamSim is 0.108, compared with 0.171 for Latent JSCC and 0.306 for ARPC; the corresponding CLIP-I values are 0.873, 0.789, and 0.713. Its PSNR exceeds ARPC by 3.26–3.97 dB across the three operating points. SwinJSCC and BPG achieve higher PSNR than NeST-Com throughout. Latent JSCC also has higher PSNR at 4 and 10 dB: 19.66 versus 17.45 dB and 20.80 versus 20.01 dB, respectively. For LPIPS, Latent JSCC leads at 4 dB, the two methods are close at 10 dB (0.1733 versus 0.1715; paired CI includes zero), and NeST-Com leads at 19 dB. On the matched 100-image subset, replacing ARPC's original sampling ($\tau = 0.5$) with argmax or $\tau = 0.25$ changes mean PSNR by at most approximately 0.15 dB, while both alternatives worsen LPIPS and DreamSim at 4 dB; these tested decoding choices therefore do not account for the multi-dB PSNR gap.

Fig. 3 illustrates the 10 dB reconstructions. NeST-Com and Latent JSCC preserve the pastry arrangement and the dog's silhouette, with clearer pastry folds and facial fur in the NeST-Com examples. SwinJSCC smooths detail, BPG introduces visible blocks, and ARPC produces texture with changes in shape and appearance. The metrics also distinguish the two leading examples: for the dog, Latent JSCC has lower LPIPS (0.203 versus 0.233), whereas NeST-Com has lower DreamSim (0.114 versus 0.166).

### C. Effect of Within-Scale Endpoints

We compare WHOLE, which permits complete-scale endpoints, with PARTIAL, which additionally permits the quarter-scale endpoints in (4). The tokenizer, prior, entropy coder, and image decoder are shared. The two endpoint sets are calibrated separately and select the same MCS at each tested SNR. Table I reports PARTIAL minus WHOLE on common500.

At 10 dB, finer endpoints raise utilization from 77.70% to 95.21%, improve PSNR by 0.701 dB, and reduce DreamSim by 0.0240. Across all three SNRs, PSNR improves by 0.32–0.70 dB, with positive CIs, and DreamSim decreases with negative CIs. The common model components identify endpoint granularity and its interaction with actual-length fitting as the source of these gains: additional boundary tokens can be retained when the next complete scale would overflow the frame.

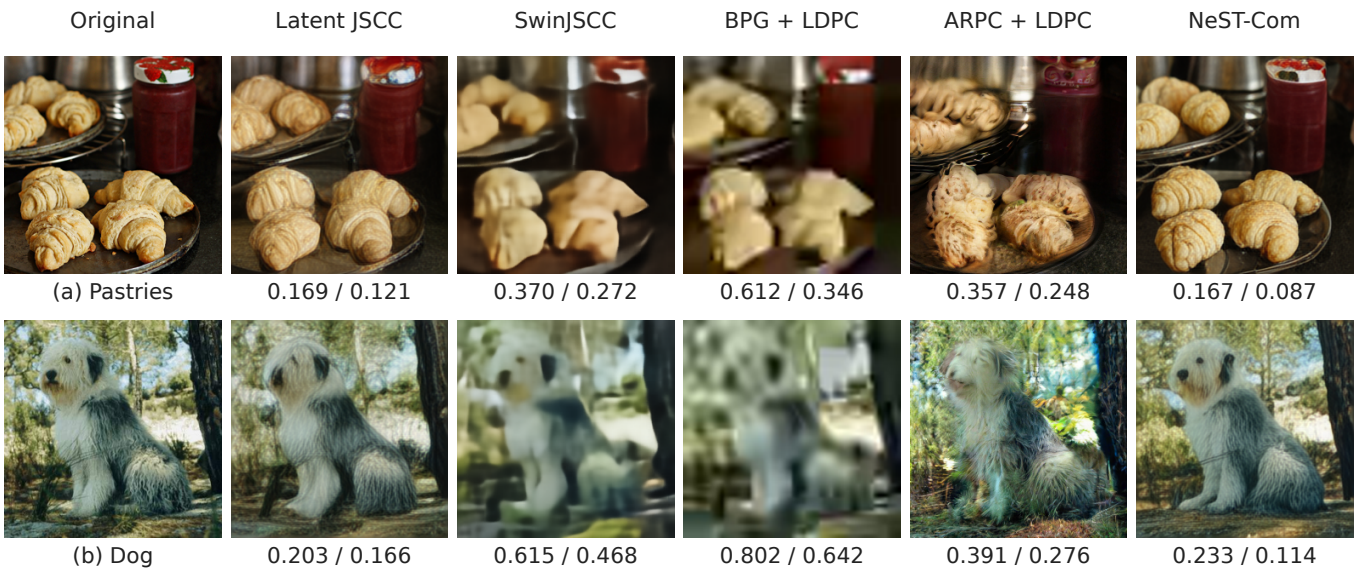



Fig. 3. Qualitative reconstructions at $N = 1024$ and 10 dB: (a) pastries and (b) a dog. Values below each reconstruction denote LPIPS-Alex↓ / DreamSim↓ for the displayed realization.

TABLE I
WITHIN-SCALE ENDPOINT GAINS ON COMMON500

| SNR | $\eta$ (%), W/P | $\Delta$PSNR (dB) | $\Delta$DreamSim |
|---|---|---|---|
| 4 dB | 73.81/93.08 | $+0.616$ | $-0.0207$ |
| | | $[0.564, 0.673]$ | $[-0.0233, -0.0183]$ |
| 10 dB | 77.70/95.21 | $+0.701$ | $-0.0240$ |
| | | $[0.637, 0.766]$ | $[-0.0270, -0.0212]$ |
| 19 dB | 85.24/94.73 | $+0.319$ | $-0.0030$ |
| | | $[0.290, 0.349]$ | $[-0.0037, -0.0024]$ |

W/P: WHOLE/PARTIAL. Differences are PARTIAL minus WHOLE; brackets give source-paired 95% CIs.

TABLE II
NEST-COM MINUS ARPC AT $N = 2048$

| Condition | $\Delta$PSNR (dB) | $\Delta$DreamSim |
|---|---|---|
| 4 dB, adaptive | $+3.779$ | $-0.2658$ |
| | $[3.493, 4.082]$ | $[-0.2933, -0.2386]$ |
| 10 dB, adaptive | $+3.877$ | $-0.1174$ |
| | $[3.688, 4.079]$ | $[-0.1309, -0.1049]$ |
| 19 dB, full | $+1.806$ | $-0.0086$ |
| | $[1.742, 1.871]$ | $[-0.0119, -0.0052]$ |

Source-paired 95% CIs over 100 images. Full denotes successful delivery of both methods' complete representations.

### D. Performance With a Larger Symbol Budget

The $N = 2048$ experiment examines whether ARPC's gap is explained by its full representation not fitting at $N = 1024$. Each image uses two 1024-symbol subframes with their own protected headers. Table II compares adaptive configurations at 4 and 10 dB and complete-representation transmission at 19 dB, using 100 images and three channel realizations per image. All 300 receptions per method succeed in each reported condition.

When both complete representations are delivered at 19 dB, NeST-Com retains a 1.806 dB PSNR advantage and lower DreamSim. ARPC has lower DISTS [14]: the NeST-Com minus ARPC difference is $+0.0054$, with 95% CI $[+0.0024, +0.0086]$. Representation and decoding therefore also affect quality, with different perceptual criteria favoring different reconstructions. The full-system gap includes these effects alongside the endpoint gains isolated in Table I.

TABLE III
MEAN PROCESSING TIME AT $N = 1024$, 10 DB (MS/IMAGE)

| Method | TX | RX | Total |
|---|---|---|---|
| NeST-Com | 220.89 | 398.82 | 619.71 |
| Latent JSCC | 17.97 | 24.51 | 42.48 |
| SwinJSCC | 24.81 | 19.89 | 44.69 |
| BPG + LDPC | 2269.87 | 23.15 | 2293.01 |
| ARPC + LDPC | 719.12 | 801.18 | 1520.30 |

### E. Computational Cost

Table III reports software processing time at 10 dB on an RTX PRO 6000 Blackwell GPU with six assigned CPU cores. Measurements use batch size one, 16 development images, three warmups, and three measured repetitions per image. TX and RX cover source handling, channel-code processing, and reconstruction; radio transmission time and model initialization are excluded. BPG's TX includes the adaptive resolution/quantization search.

NeST-Com takes 619.71 ms per image, compared with 42–45 ms for the continuous JSCC baselines, 1520.30 ms for ARPC, and 2293.01 ms for adaptive BPG. Its 398.82 ms receiver cost reflects local probability reconstruction and next-scale completion. These results expose the processing cost accompanying its perceptual and image-feature similarity gains.

## IV. Conclusion

We presented NeST-Com, which fits next-scale token prefixes to a fixed-symbol digital frame using within-scale endpoints and actual terminated lengths. A shared prior supports entropy decoding and reconstruction constrained by the received tokens. Under the same prior and decoder, finer endpoints improve PSNR by 0.32–0.70 dB and increase utilization from approximately 78% to 95% at 10 dB. The five-method comparison shows favorable DreamSim and CLIP-I performance, together with SNR-dependent fidelity tradeoffs and higher processing cost than continuous JSCC.

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